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From Foundations to Applications

Matplotlib

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Matplotlib is known as the “Grandfather” of plotting and visualization libraries.

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- Matplotlib is heavily inspired by the plotting functions of the MatLab programming language.

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- It allows for the creation of almost any plot type and heavy customization.

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- This ability to heavily customize a plot comes at a trade-off for beginners, since it can be confusing to learn the Matplolib syntax at first.

- This is mainly due to the fact that there are actually two separate approaches to creating plots, functional based methods and OOP based methods.

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Topics Covered

- Matplotlib Basics and Functions
- Matplotlib Figures
- Matplotlib Subplots
- Matplotlib Styling
- Exercise Questions and Solutions

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Specialized plot types such as histograms won't be covered with matplotlib, since we will later learn how to use seaborn to easily create statistical plots.

It is important to learn matplotlib first however, since seaborn builds directly off of Matplotlib.

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Two main goals with Matplotlib:

- Be able to plot out a functional relationship:
  - y = 2x
- Be able to plot out a relationship between raw data points:
  - x = [1,2,3,4]
  - y = [2,4,6,8]

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Let’s get started!

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Matplotlib Basics

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plot calls:

- **plt.plot(x,y)**

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Simple to use yet they don’t allow for very high degrees of control.

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We recommend using these simple plt.plot() calls for quickly visualizing relationships and data.

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NumPy

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Almost every data science library we learn about in this course is built using NumPy!

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Let’s explore what this section will cover and why it is important.

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1. Understand NumPy
2. Create arrays with NumPy
3. Retrieve information from a NumPy array through slicing and indexing.
4. Learn basic NumPy operations
5. Test NumPy skills with exercise questions.

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What is NumPy?

- Python library for creating N-dimensional arrays
- Built-in linear statistical distributions, algebra, trigonometric, and random number capabilities

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Why NumPy?

- While NumPy structures look similar to standard Python lists, they are **much** more efficient!
- The broadcasting capabilities helps in quickly applying functions to our data sets.

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Numpy Arrays

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- Let’s explore how to create NumPy arrays
  - Transforming standard list
  - Built-in functions
  - Generating random data
- We’ll also discuss some key attributes of NumPy arrays.

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Numpy
Indexing and Selection

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- Grabbing a element
- Grabbing a slice of elements
- Broadcasting selections
- Conditional selection
- Indexing and selection in 2-D

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Numpy Operations

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- A key feature of NumPy is its ability to perform arithmetic on large arrays on an **element by element** basis.
- Let’s learn how this works with multiple arrays, as well as how we can apply universal array functions.